

شبیه سازی سوال چهارم

آرمین برزگری

بخش الف) بدون محدودیت:

کد:

```
clear; clc;

mpc.version = '2';
mpc.baseMVA = 100;

% اطلاعات باس ها
% [bus_i type Pd Qd Gs Bs area Vm Va baseKV zone Vmax Vmin]
mpc.bus = [
    1 3 0 0 0 0 1 1.04 0 138 1 1.1 0.9;
    2 2 0 0 0 0 1 1.04 0 138 1 1.1 0.9;
    3 1 100 50 0 0 1 1.00 0 138 1 1.1 0.9;
    4 1 30 10 0 0 1 1.00 0 138 1 1.1 0.9;
];

% (Q بدون محدودیت) اطلاعات ژنراتورها
% [bus Pg Qg Qmax Qmin Vg mBase status Pmax Pmin]
mpc.gen = [
    1 0 0 9999 -9999 1.04 100 1 9999 -9999; % Slack
    2 50 0 9999 -9999 1.04 100 1 9999 -9999; % Gen 2
];
```

% اطلاعات خطوط

% [fbus tbus r x b rateA rateB rateC ratio angle status angmin angmax]

mpc.branch = [

1 2 0 0.20 0 0 0 0 0 0 1 -360 360;

1 3 0 0.40 0 0 0 0 0 0 1 -360 360;

2 3 0 0.50 0 0 0 0 0 0 1 -360 360;

2 4 0 0.50 0 0 0 0 0 0 1 -360 360;

3 4 0 0.25 0 0 0 0 0 0 1 -360 360;

];

disp('=====');

disp(' PART A: Power Flow WITHOUT Q Limits on Gen 2 ');

disp('=====');

% اجرای پخش بار

results_A = runpf(mpc);

V2_A = results_A.bus(2, 8);

Qg2_A = results_A.gen(2, 3);

fprintf('\n>>> RESULTS FOR PART A <<<\n');

fprintf('Voltage at Bus 2: %.4f pu\n', V2_A);

fprintf('Reactive Power Generated by Gen 2: %.4f MVAR (%.4f pu)\n\n', Qg2_A,
Qg2_A/mpc.baseMVA);

خروجی:

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PART A: Power Flow WITHOUT Q Limits on Gen 2

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MATPOWER Version 8.1, 12-Jul-2025

Power Flow -- AC-polar-power formulation

Newton's method converged in 4 iterations.
PF successful

Converged in 0.08 seconds

```
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===
|      System Summary
|
=====
===
```

How many?		How much?	P (MW)	Q (MVar)

-				
Buses	4	Total Gen Capacity	19998.0	-19998.0 to
19998.0				
Generators	2	On-line Capacity	19998.0	-19998.0 to
19998.0				
Committed Gens	2	Generation (actual)	130.0	99.5
Loads	2	Load	130.0	60.0
Fixed	2	Fixed	130.0	60.0
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	0	Shunt (inj)	-0.0	0.0
Branches	5	Losses ($I^2 * Z$)	-0.00	39.47
Transformers	0	Branch Charging (inj)	-	0.0
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
	-----	-----
Voltage Magnitude	0.909 p.u. @ bus 3	1.040 p.u. @ bus 1
Voltage Angle	-13.94 deg @ bus 3	0.00 deg @ bus 1
P Losses ($I^2 * R$)	-	0.00 MW @ line 1-2
Q Losses ($I^2 * X$)	-	18.20 MVar @ line 1-3

```
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|      Bus Data
|
=====
===
```

Bus #	Voltage		Generation		Load	
	Mag (pu)	Ang (deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
1	1.040	0.000*	80.00	41.52	-	-
2	1.040	-2.447	50.00	57.95	-	-
3	0.909	-13.935	-	-	100.00	50.00
4	0.929	-13.014	-	-	30.00	10.00
Total:			130.00	99.47	130.00	60.00

```
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===
|      Branch Data
|
=====
```

```

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===
Brnch      From   To   From Bus Injection   To Bus Injection   Loss (I^2 * Z)
#          Bus   Bus   P (MW)  Q (MVar)  P (MW)  Q (MVar)  P (MW)  Q
(MVar)
-----
--
1          1     2    23.09   0.49    -23.09   0.49     0.000
0.99
2          1     3    56.91  41.02   -56.91  -22.82     0.000
18.20
3          2     3    37.66  31.04   -37.66  -20.03     0.000
11.01
4          2     4    35.43  26.42   -35.43  -17.39    -0.000
9.03
5          3     4    -5.43  -7.15    5.43    7.39     0.000
0.24
-----
--
Total:      -0.000
39.47

```

>>> RESULTS FOR PART A <<<

Voltage at Bus 2: 1.0400 pu

Reactive Power Generated by Gen 2: 57.9548 MVAR (0.5795 pu)

اندازه ولتاژ باس 2 برابر با 1.04 پریونیت است

بخش ب) با اعمال محدودیت ذکر شده:

کد:

```
clear; clc;
```

```
mpc.version = '2';
```

```
mpc.baseMVA = 100;
```

% اطلاعات باس ها

```
% [bus_i type Pd Qd Gs Bs area Vm Va baseKV zone Vmax Vmin]
```

```
mpc.bus = [
```

```
1 3 0 0 0 0 1 1.04 0 138 1 1.1 0.9;
```

```
2 2 0 0 0 0 1 1.04 0 138 1 1.1 0.9;
```

```
3 1 100 50 0 0 1 1.00 0 138 1 1.1 0.9;
```

```

4 1 30 10 0 0 1 1.00 0 138 1 1.1 0.9;

];

% (Qmax=25 و Qmin=-20 اعمال محدودیت) اطلاعات ژنراتورها
% [bus Pg Qg Qmax Qmin Vg mBase status Pmax Pmin]
mpc.gen = [
1 0 0 9999 -9999 1.04 100 1 9999 -9999; % Slack
2 50 0 25 -20 1.04 100 1 9999 -9999; % Gen 2 Limits applied
];

% اطلاعات خطوط
% [fbus tbus r x b rateA rateB rateC ratio angle status angmin angmax]
mpc.branch = [
1 2 0 0.20 0 0 0 0 0 0 1 -360 360;
1 3 0 0.40 0 0 0 0 0 0 1 -360 360;
2 3 0 0.50 0 0 0 0 0 0 1 -360 360;
2 4 0 0.50 0 0 0 0 0 0 1 -360 360;
3 4 0 0.25 0 0 0 0 0 0 1 -360 360;
];

disp('=====');
disp(' PART B: Power Flow WITH Q Limits on Gen 2 ');
disp('=====');

% فعال کردن اعمال محدودیت های توان راکتیو در تنظیمات متپاور
mpopt = mpooption('pf.enforce_q_lims', 1);

```

اجرای پخش بار با تنظیمات جدید %

```
results_B = runpf(mpc, mpopt);
```

```
V2_B = results_B.bus(2, 8);
```

```
Qg2_B = results_B.gen(2, 3);
```

```
fprintf('\n>>> RESULTS FOR PART B <<<\n');
```

```
fprintf('Voltage at Bus 2: %.4f pu\n', V2_B);
```

```
fprintf('Reactive Power Generated by Gen 2: %.4f MVAR (%.4f pu)\n', Qg2_B,  
Qg2_B/mpc.baseMVA);
```

بررسی تغییر نوع باس %

```
bus2_type_after = results_B.bus(2, 2);
```

```
if bus2_type_after == 1
```

```
    disp('Notice: Bus 2 reached its Q limit and was converted from PV to PQ bus.');
```

```
end
```

خروجی:

```
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PART B: Power Flow WITH Q Limits on Gen 2
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MATPOWER Version 8.1, 12-Jul-2025
Power Flow -- AC-polar-power formulation

Newton's method converged in 4 iterations.
PF successful
Gen 2 at upper Q limit, converting to PQ bus

Newton's method converged in 3 iterations.
PF successful

Converged in 0.04 seconds
=====
===
|      System Summary
|
=====
===
```

How many?		How much?	P (MW)	Q (MVar)
-----		-----	-----	-----
-				
Buses	4	Total Gen Capacity	19998.0	-10019.0 to
10024.0				
Generators	2	On-line Capacity	19998.0	-10019.0 to
10024.0				
Committed Gens	2	Generation (actual)	130.0	104.8
Loads	2	Load	130.0	60.0
Fixed	2	Fixed	130.0	60.0
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	0	Shunt (inj)	-0.0	0.0
Branches	5	Losses ($I^2 * Z$)	0.00	44.77
Transformers	0	Branch Charging (inj)	-	0.0
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
-----	-----	-----
Voltage Magnitude	0.871 p.u. @ bus 3	1.040 p.u. @ bus 1
Voltage Angle	-14.88 deg @ bus 3	0.00 deg @ bus 1
P Losses ($I^2 * R$)	-	0.00 MW @ line 1-2
Q Losses ($I^2 * X$)	-	22.35 MVar @ line 1-3

```

=====
===
|      Bus Data
|
=====
=====

```

Bus	Voltage		Generation		Load	
#	Mag(pu)	Ang(deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
-----	-----	-----	-----	-----	-----	-----
1	1.040	0.000*	80.00	79.77	-	-
2	0.987	-2.443	50.00	25.00	-	-
3	0.871	-14.880	-	-	100.00	50.00
4	0.884	-13.974	-	-	30.00	10.00
			-----	-----	-----	-----
Total:			130.00	104.77	130.00	60.00

```

=====
===
|      Branch Data
|
=====
=====

```

Brnch	From	To	From Bus	Injection	To Bus	Injection	Loss ($I^2 * Z$)	
#	Bus	Bus	P (MW)	Q (MVar)	P (MW)	Q (MVar)	P (MW)	Q
-----	-----	-----	-----	-----	-----	-----	-----	-----
--								
1	1	2	21.87	28.15	-21.87	-25.80	0.000	
2.35								
2	1	3	58.13	51.63	-58.13	-29.27	0.000	
22.35								

